# Spectrally-consistent neural surrogate for real-time ultrasound pressure-field prediction in heterogeneous breast tissue

**Background.** Patient-specific HIFU planning requires fast pressure-field simulation in heterogeneous tissue. Full-wave k-Wave delivers reference accuracy but costs **~60 s per 316 × 256 / 1 MHz configuration** on an RTX 4070 — an order of magnitude too slow for the sub-second feedback clinicians need while sweeping the focal target. Frequency-domain Helmholtz, ray tracing and lookup tables each fail on a different axis (transient information, refraction tail, or per-patient adaptivity), defining the engineering slot for a learned forward surrogate that is on-the-go and GPU-light.

**Method.** We trained a 2-D Fourier Neural Operator (FNO) and a matched 2-D U-Net baseline on **1 000 simulations** drawn from the recently released **OpenBreastUS** dataset (Zeng *et al.* 2025) — anatomically realistic sound-speed, density and attenuation maps derived from real breast MRI, chosen for *in vivo* fat / fibroglandular impedance contrast that synthetic phantoms cannot reproduce. Identical 70 / 15 / 15 split (seed 0) across all backbones; combined `LpLoss + 0.3·H1Loss` objective; per-channel z-scored inputs; `log1p(p_max)` z-scored targets recovered to Pa via `expm1`.

**Results.** FNO achieves **test relative L2 = 0.097**, against U-Net's **0.264** (**2.7**× lower error) and a ConvNeXt-2D ablation's **0.99** (an order of magnitude worse). Inference completes in **~8 ms / sample** on the same RTX-4070 — a **~7 500× speed-up** over k-Wave at matched accuracy on the test split. Validation loss scaled 0.81 → 0.39 from 200 → 1 000 samples, indicating substantial headroom on the full 8 000-phantom set. Visual inspection confirms FNO recovers the refraction tail that U-Net smears, consistent with FFT-based representations matching wave-equation structure.

**Significance.** To our knowledge this is the first application of OpenBreastUS to ultrasound treatment-planning surrogates (the original release targets ultrasound CT reconstruction). Prior neural-operator work (TUSNet 2022, DeepTFUS 2023, Stanziola 2023) used non-breast tissue or synthetic phantoms; the **spectrally-consistent backbone advantage** we report (FNO vs generic CNNs at matched parameter budgets) is a heterogeneous-tissue-specific finding not previously established and a clinically relevant building block for real-time HIFU planning.

*Keywords*: HIFU, Fourier Neural Operator, k-Wave, OpenBreastUS, real-time treatment planning, wave equation, breast ultrasound.